System Administration and Maintenance

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5. Software configurations on computer Networks

* The process of assigning:
  - Network settings
  - Policies
  - Flows
  - Controls

Learning Outcomes

* **Demonstrate an understanding of the components of a computer network.**
* **Implement a computer network for an existing requirement.**
* **Identify and explain network configurations.**

Topics Covered

- Network configuration concepts
- IP addressing and subnet configuration
- DHCP and DNS
- Network interface configuration
- Routing configuration
- Network services
- Security policies and controls
- Practical network configuration
- Network troubleshooting

What is Network Configuration?

Definition

* **Network configuration** is the process of setting up and managing the software parameters that allow network devices and computers to communicate correctly.

* **It includes:**
  - IP addressing
  - Subnet masks
  - Default gateways
  - DNS servers
  - DHCP settings
  - Routing
  - VLAN configuration
  - Network protocols
  - Security policies
  - Access controls
  - Firewall rules

* **Hardware + Software Configuration = Functional Network**

Why is Network Configuration Important?

* **Proper configuration ensures:**
  - Devices can communicate.
  - Users can access network resources.
  - Internet connectivity is available.
  - Network traffic follows appropriate paths.
  - Unauthorized access is restricted.
  - Network resources are efficiently shared.
  - Problems can be identified and resolved.

* Eg:
  - A computer may have:

    NIC + Ethernet cable + Switch

  - but without appropriate **IP configuration**, communication may not work correctly.

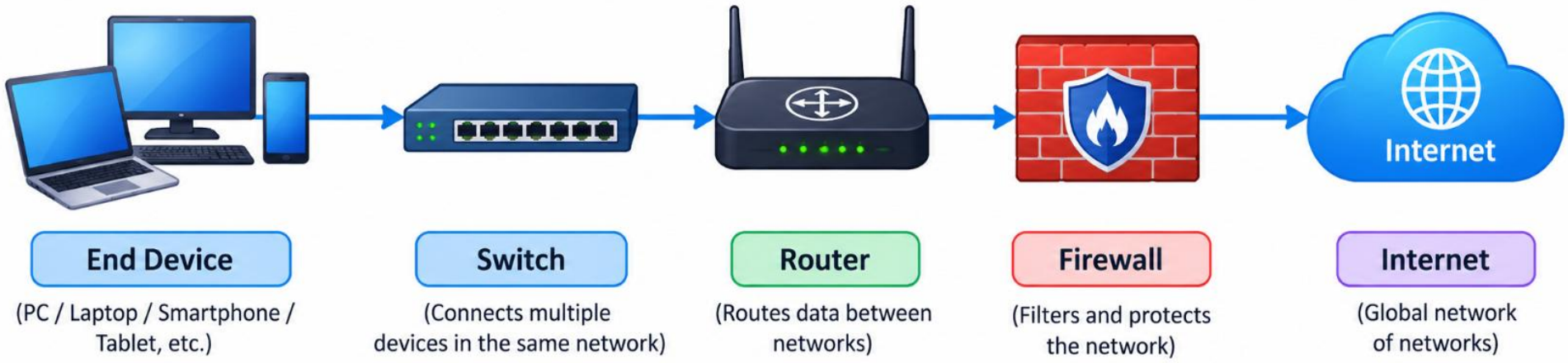
Components Involved in Network Configuration

* **Hardware Components**
  - Computers / Hosts
  - Network Interface Cards
  - Switches
  - Routers
  - Wireless Access Points
  - Servers
  - Firewalls
  - Network cables

* **Software Components**
  - Operating systems
  - Network drivers
  - TCP/IP stack
  - DHCP
  - DNS
  - Routing software
  - Firewall software
  - Network management tools

Network Configuration Model

* **Software configuration determines:**
  - How devices identify themselves
  - Where traffic should go
  - Which services can be accessed
  - Which traffic should be blocked
  - How names are resolved



IP Network Configuration

* **Refer Practical Slides – IP Addressing and IP Classes**

* **IP Address**
  - An IP address uniquely identifies a device on an IP network.

* **IPv4 Example**
  192.168.1.10

* **IPv4 address consists of:**
  - Network portion
  - Host portion

* **Example Network**
  - **Network:** 192.168.1.0/24

* **Possible hosts:**
  - 192.168.1.1
  - 192.168.1.2
  - 192.168.1.10
  - 192.168.1.100

Main IPv4 Configuration Parameters

* **Refer Practical Slides – IP Addressing and IP Classes**

Parameter	Example
IP Address	192.168.1.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	8.8.8.8

Purpose

IP Address → Identifies the device

Subnet Mask → Defines the network

Default Gateway → Provides access to other networks

DNS Server → Converts domain names to IP addresses

Main IPv4 Configuration Parameters

Parameter	**Example**
IP Address	192.168.1.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	8.8.8.8

Purpose

* **IP Address** → Identifies the device on the network (Server, Hosts, Routers, Switches and etc..)
* **Subnet Mask** → Defines the network
* **Default Gateway** → Provides access to other networks
* **DNS Server** → Converts domain names to IP addresses

Static IP Configuration

* The network administrator manually assigns network settings.

* Eg:
  - IP Address : 192.168.10.25
  - Subnet Mask : 255.255.255.0
  - Default Gateway : 192.168.10.1
  - DNS Server : 8.8.8.8

* **Suitable for:**
  - Servers
  - Network printers
  - Routers
  - Network management devices
  - CCTV/NVR systems
  - Infrastructure devices

* **Advantage**
  - Predictable IP address.

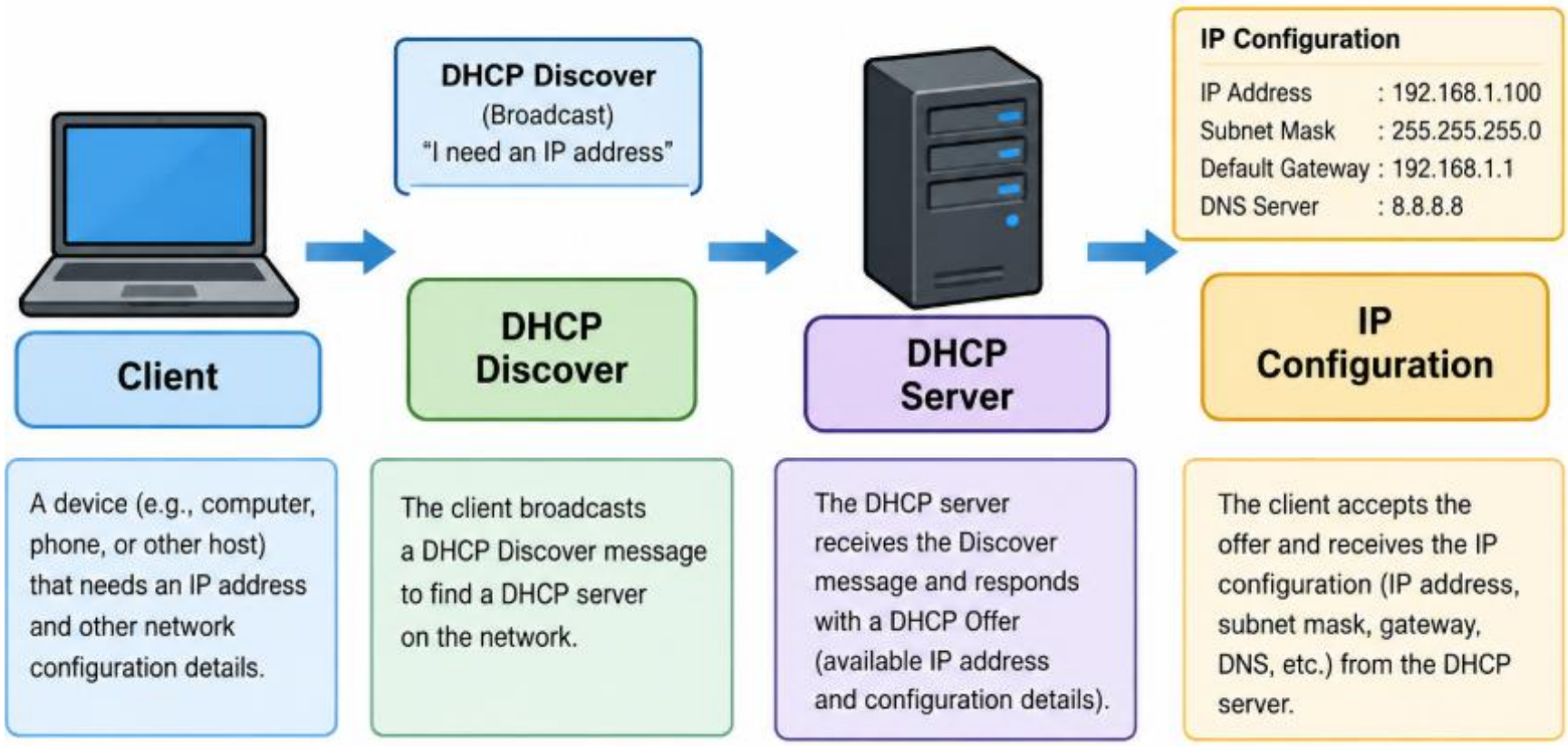
* **Disadvantage**
  - Manual administration is required.

DHCP – Dynamic Host Configuration Protocol

* DHCP automatically provides network configuration to clients.

* **DHCP can provide:**
  - IP address
  - Subnet mask
  - Default gateway
  - DNS server
  - Lease duration / time

* Eg:


DHCP Discover
(Broadcast)
"I need an IP address"
IP Configuration
IP Address : 192.168.1.100
Subnet Mask : 255.255.255.0
Default Gateway : 192.168.1.1
DNS Server : 8.8.8.8
Client
DHCP Discover
DHCP Server
IP Configuration
A device (e.g., computer, phone, or other host) that needs an IP address and other network configuration details.
The client broadcasts a DHCP Discover message to find a DHCP server on the network.
The DHCP server receives the Discover message and responds with a DHCP Offer (available IP address and configuration details).
The client accepts the offer and receives the IP configuration (IP address, subnet mask, gateway, DNS, etc.) from the DHCP server.

DHCP Process

* **DORA Process**

D – Discover

- Client searches for a DHCP server.

O – Offer

- DHCP server offers an IP address.

R – Request

- Client requests the offered address.

A – Acknowledge

- Server confirms the allocation.

Static vs Dynamic Configuration

Feature	Static	DHCP
Assignment	Manual	Automatic
Administration	Higher	Lower
Suitable for servers	Yes	Usually reservation
Suitable for clients	Sometimes	Yes
Configuration errors	More likely	Reduced
Central management	Limited	Strong

DNS Configuration

* **DNS – Domain Name System**

DNS translates human-readable domain names into IP addresses.

* **Why DNS is required**

* Users remember : **www.google.com**
* Rather than : **IP address**

www.example.com

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DNS Server

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93.184.216.34

DNS Configuration Parameters

* Preferred DNS server
* Alternate DNS server
* DNS search domain
* DNS suffix

* Eg:

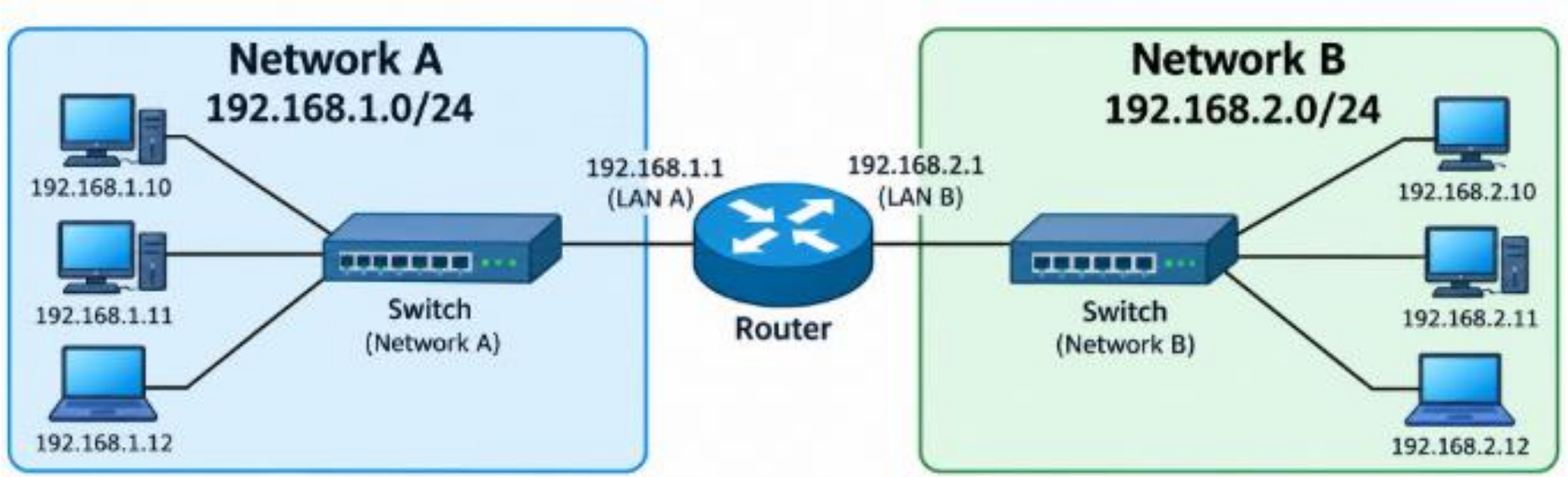
* Preferred DNS : 8.8.8.8
* Alternate DNS : 1.1.1.1

Organizational DNS

* Organizations may use internal DNS servers for:
  - server01.**company.local**
  - printer01.**company.local**
  - mail.**company.local**

Static Routing

* **Static Route**
* A route manually configured by the administrator.



* **Advantages**
  - Simple
  - Predictable
  - No routing protocol overhead

* **Disadvantages**
  - Manual configuration
  - Difficult to manage in large networks
  - Changes require administrator intervention

Dynamic Routing

* Dynamic routing protocols automatically exchange routing information.
* **Eg:**
  - RIP **(Routing Information Protocol)**
  - OSPF **(Open Shortest Path First)**
  - EIGRP **(Enhanced Interior Gateway Routing Protocol)**
  - IS-IS **(Intermediate System to Intermediate System)**
  - BGP **(Border Gateway Protocol)**

* https://diog0net0sec.medium.com/dynamic-routing-protocols-rip-igrp-eigrp-ospf-is-is-egp-and-bgp-explained-2e93fa87a021

* **Benefits**
  - Automatic route updates
  - Suitable for larger networks
  - Can adapt to network changes

* **Eg:** **OSPF** is widely used within enterprise networks.

Linux Network Configuration

Common commands

* Displays IP addresses.

  ip addr

* Displays routing table.

  ip route

* Displays network interfaces.

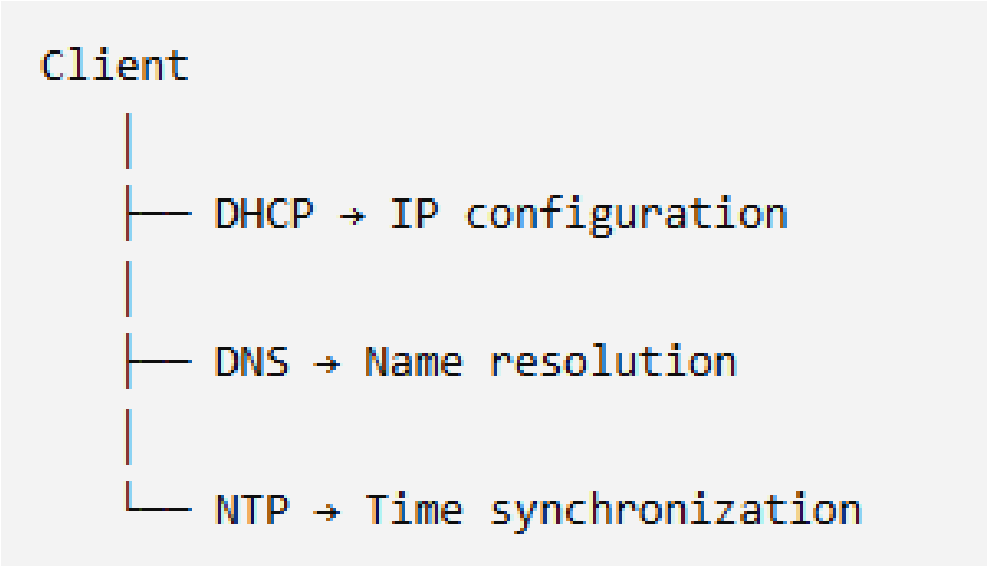
  ip link

* Testing

  Ping 192.168.1.1

Network Services Configuration

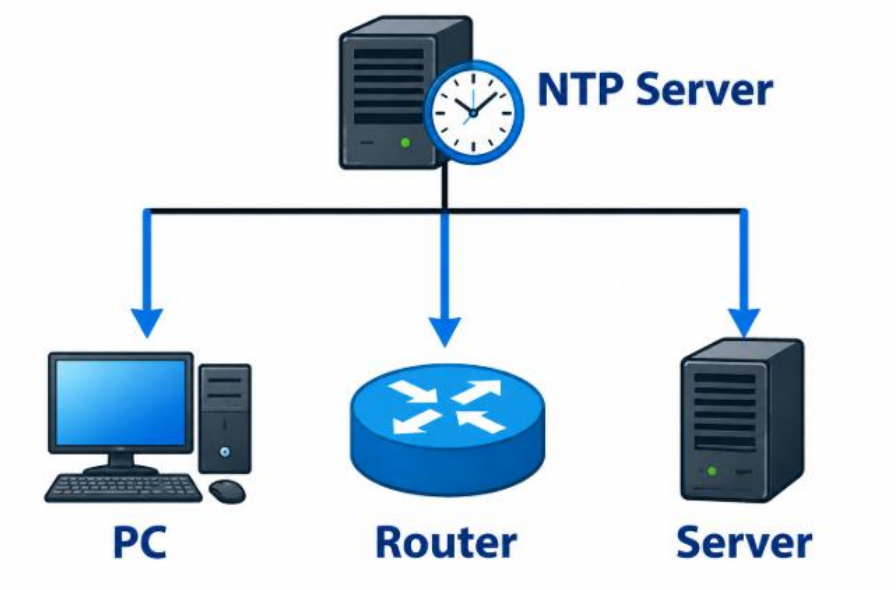
* Network configuration commonly involves:
  - DHCP
  - DNS
  - Web servers
  - File servers
  - Print servers
  - Email servers
  - Authentication servers
  - NTP
  - Proxy servers



NTP Configuration

* **NTP – Network Time Protocol**
  Synchronizes clocks across network devices.

* **Why is accurate time important?**
  - Security logs
  - Authentication
  - Digital certificates
  - Event correlation
  - Troubleshooting
  - Database systems



Network Security Configuration

* Network configuration must include security controls.

* **Common controls**
  - Firewall rules
  - Access Control Lists
  - Authentication
  - Password policies
  - Port security
  - VLAN segmentation
  - VPN configuration
  - Wireless security
  - Network monitoring

* **Goal :** Allow legitimate traffic and prevent unauthorized access.

Firewall Configuration

* **Firewall -** A firewall controls network traffic according to predefined rules.

Source	Destination	Action
192.168.1.0/24	Web Server	ALLOW
Internet	Internal DB	DENY
Admin PC	Server SSH	ALLOW

➢ **Firewall decisions –**

- Allow
- Deny
- Reject
- Log

Access Control Lists (ACL)

* **ACLs can control:**
  - Source IP
  - Destination IP
  - Protocol
  - Port number
  - Direction

- Eg:

Allow:
192.168.10.0/24 → Server VLAN

Deny:
Guest VLAN → Server VLAN

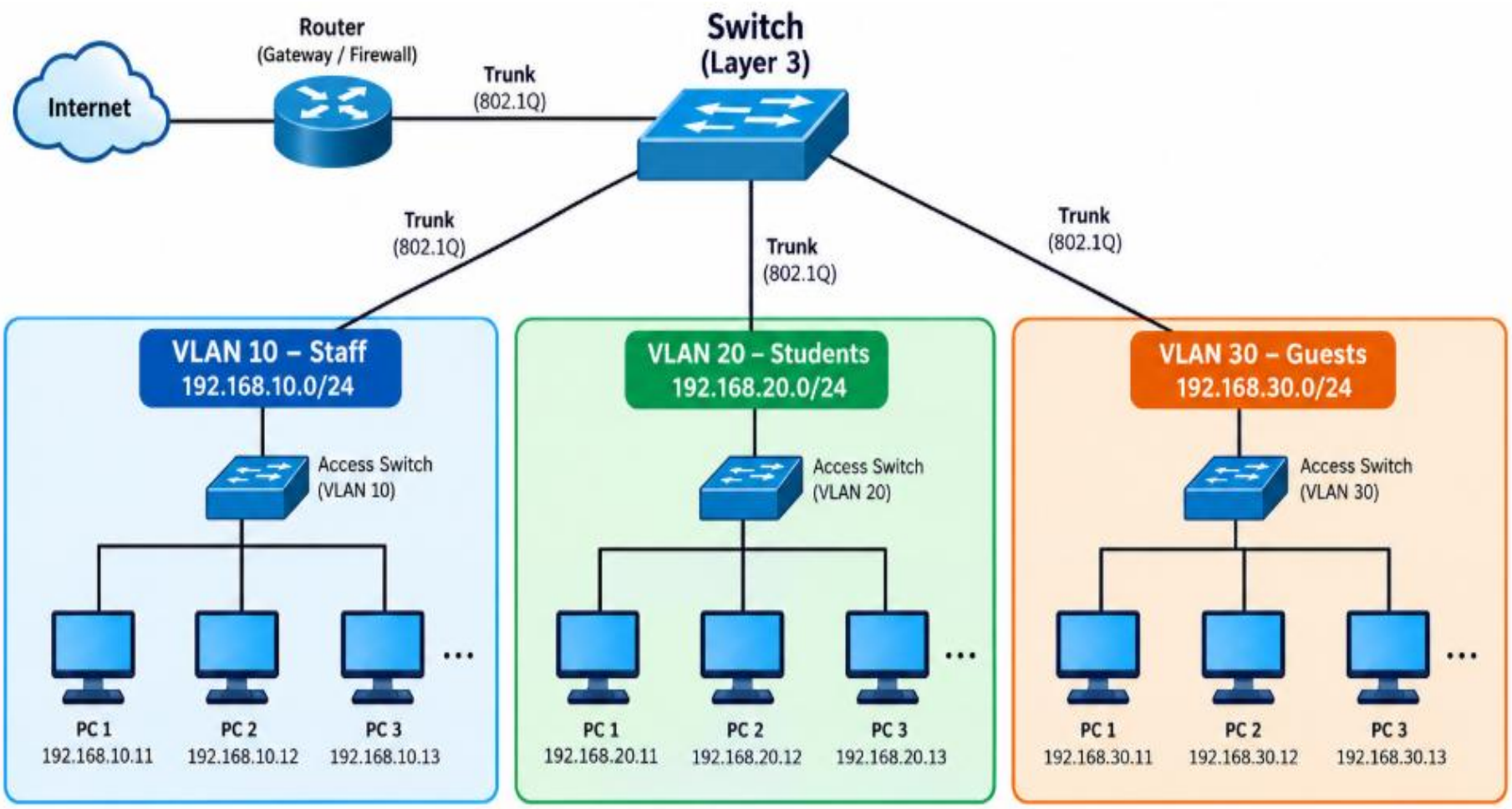
VLAN Configuration

* **Virtual Local Area Network**

A VLAN logically separates devices within a switched network.

* **Benefits**

- Logical segmentation
- Improved security
- Reduced broadcast domains
- Easier network management



Legend

- Router (Gateway / Firewall)
- Layer 3 Switch (Core)
- Switch (Access)
- PC / Laptop (End Device)

VLANs and IP Ranges

VLAN 10 (Staff)	: 192.168.10.0/24
VLAN 20 (Students)	: 192.168.20.0/24
VLAN 30 (Guests)	: 192.168.30.0/24

Note: VLAN 10, 20 and 30 represent separate IP networks (subnets), not the quantity of staff, students or guests.

VLAN Example

VLAN	Department	Network
VLAN 10	Administration	192.168.10.0/24
VLAN 20	Academic	192.168.20.0/24
VLAN 30	Students	192.168.30.0/24
VLAN 40	Guest	192.168.40.0/24

- **Requirement**

Students should not automatically have access to administrative systems.

- **Solution**

VLAN + Routing + ACL/Firewall

Network Policies

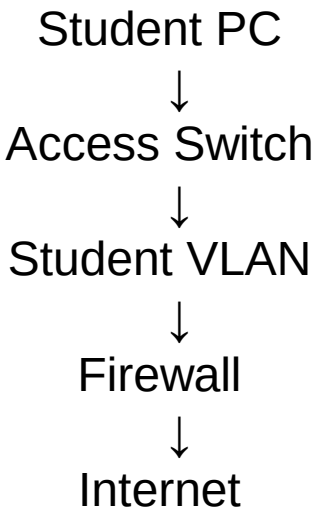
* A network policy defines how network resources should be used and protected.

* **Examples**
  - Internet usage policy
  - Password policy
  - Wi-Fi access policy
  - Remote access policy
  - Firewall policy
  - BYOD policy
  - Data access policy
  - Guest network policy

* **Example**
  - **Students → Internet : ALLOW**
  - **Students → Administration Database : DENY**

Network Traffic Flow

* **Traffic flow determines:**
  - Source
  - Destination
  - Route
  - Protocol
  - Port
  - Security controls



Note :Security controls are applied at appropriate points.

Configuration Checklist

End Device

- ☐ NIC enabled
- ☐ IP address configured
- ☐ Subnet mask configured
- ☐ Gateway configured
- ☐ DNS configured

Switch

- ☐ Ports configured
- ☐ VLAN configured
- ☐ Management IP configured

Router

- ☐ Interfaces configured
- ☐ Routing configured
- ☐ NAT configured if required

Firewall

- ☐ Security rules configured
- ☐ Unnecessary traffic blocked
- ☐ Logging enabled where appropriate

Network Configuration Best Practices

* **1. Plan before configuring**
  * Understand the requirements.
* **2. Use an IP addressing plan**
  * Avoid duplicate IP addresses.
* **3. Use DHCP where appropriate**
  * Reduce manual configuration.
* **4. Segment the network**
  * Use VLANs where appropriate.
* **5. Apply security controls**
  * Follow the principle of least privilege.

* 6. Document everything
  * Maintain configuration records.
* 7. Back up configurations
  * Allow recovery after failures.
* 8. Monitor the network
  * Identify problems early.

Common Configuration Errors

* **Error 1** - Duplicate IP addresses
* **Error 2** - Incorrect subnet mask
* **Error 3** - Wrong default gateway
* **Error 4** - Incorrect DNS server
* **Error 5** - Incorrect VLAN
* **Error 6** - Firewall blocking required traffic
* **Error 7** - Incorrect routing
* **Error 8** - DHCP scope exhausted
* **Error 9** - Incorrect wireless security configuration

Q & A